James V. Raj

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Research Interests

My research is in complex systems, statistical physics, and network science. I am particularly interested in physical learning and in topological data analysis (especially persistent homology). I also study spatial networks, such as the beam networks of scale-rich metamaterials and networks that are generated from hyperuniform point patterns.

Education

University of California, Los Angeles **Sep. 2022 – June 2027**
Ph.D. in Physics

- *Advisor: Mason A. Porter*
- *Advanced to candidacy March 2025; expected completion June 2027*

California Polytechnic State University, San Luis Obispo **Sep. 2017 – June 2021**
B.S. in Physics

Research Experience

Graduate Student Researcher **Oct. 2023 – present**
UCLA, Los Angeles, CA

- Advisor: Mason A. Porter, Department of Mathematics
- Collaborated with experimentalists in an NSF DMREF collaboration on disordered network metamaterials, generating the networks that collaborators at NC State 3D-printed and tested.
- Developed local learning rules that open spectral gaps in spring networks and studied how the structure of a network sets the time that it takes to learn.
- Developed theory that connects topological data analysis to the statistical physics of point patterns.

Graduate Student Researcher **June 2022 – June 2023**
UCLA, Los Angeles, CA

- Advisor: Shenshen Wang, Department of Physics and Astronomy
- Theoretical biophysics of adaptive immunity and evolving populations.

Postbaccalaureate Researcher **June 2021 – June 2022**
UC Santa Barbara, Materials Research Laboratory, Santa Barbara, CA

- Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
- Designed and ran phase-field simulations of reactive polymer blending under shear flow on UCSB's high-performance computing clusters; wrote the Python simulation-input and ParaView analysis scripts.

Summer Research Intern (NSF-funded) **June 2020 – Sep. 2020**
UC Santa Barbara, Future Leaders in Advanced Materials (FLAM) internship, Santa Barbara, CA

- Advisor: Glenn H. Fredrickson, Department of Chemical Engineering
- Simulated polymer membrane formation on high-performance computing clusters; wrote Python tools to quantify pore structure in SEM images and compare it with simulations.

Soft Matter Research Assistant **Jan. 2019 – June 2021**

Cal Poly, San Luis Obispo, CA

- Advisors: Hilary Jacks and Nathan Keim, Physics Department
- Wrote Python code that tests whether global statistics of particle positions encode memories in a model of non-Brownian suspensions under cyclic shear. Developed an algorithm that reads multiple stored memories from these statistics.

Nanoelectronics Research Assistant **Sep. 2018 – Aug. 2019**

Cal Poly, San Luis Obispo, CA

- Advisor: Colleen Marlow, Physics Department; includes a 2019 summer research internship
- Extended a Python simulation of carbon nanotube network field-effect transistors to model local electrostatic gating. Used it to relate the density of junctions between metallic and semiconducting nanotubes to the sensitivity of a network and to choose morphological parameters that guarantee percolation without reducing that sensitivity.

Publications

Peer-reviewed journal articles

1. R. Sengupta, M. D. Tikekar, **J. V. Raj**, K. T. Delaney, M. C. Villet, and G. H. Fredrickson, "Phase-field simulations of morphology development in reactive polymer blending," *J. Rheol.* **67**, 1 (2023). doi:10.1122/8.0000523

Preprints and papers under review

2. **J. V. Raj**, X. Sun, C. E. Maher, K. A. Newhall, and M. A. Porter, "Local geometric and transport properties of networks that are generated from hyperuniform point patterns," arXiv:2511.21082 (2025); resubmitted to *Phys. Rev. E*. Code: github.com/DMREF-networks/HuPPI-Network-Analysis.

Manuscripts in preparation

3. **J. V. Raj**, P. V. Pietrzkowski, L. Gyllingberg, M. Stern, and M. A. Porter, "Local physical learning opens global spectral gaps" (2026).
4. **J. V. Raj**, L. Gyllingberg, P. V. Pietrzkowski, M. Stern, and M. A. Porter, "Network structure controls how quickly local learning opens spectral gaps in spring networks" (2026).
5. S. Kondaveeti, **J. V. Raj**, M. A. Porter, and K. E. Daniels, "Specific modulus of 3D-printed disordered networks" (2026).
6. **J. V. Raj**, C. E. Maher, K. A. Newhall, and M. A. Porter, "Topological data analysis for statistical physics" (2026).
7. **J. V. Raj**, C. E. Maher, K. A. Newhall, and M. A. Porter, "A directional transport surrogate for stiffness in disordered beam networks" (2026).

Presentations

1. **[Contributed Talk] J. V. Raj**, L. Gyllingberg, P. V. Pietrzkowski, M. Stern, and M. A. Porter, "Network structure shapes accessible solutions in physical learning," Conference on Complex Systems (CCS 2026), Binghamton, NY, Oct. 2026.
2. **[Contributed Talk]** K. E. Daniels, S. Gat, S. Kondaveeti, C. E. Maher, M. A. Porter, **J. V. Raj**, S. Ramirez, and C. Rock, "Characterizing 3D-printed disordered metamaterials under static and oscillatory driving," APS Global Physics Summit, Denver, CO, Mar. 2026. (Presented by K. E. Daniels.)
3. **[Contributed Talk] J. V. Raj** and M. A. Porter, "Topological and transport properties of spatially embedded networks generated by hyperuniform point patterns," APS March Meeting, Anaheim, CA, Mar. 2025.
4. **[Contributed Talk] J. V. Raj**, J. Garcia, and G. H. Fredrickson, "Understanding how the glass transition concentration affects polymer membrane microstructure," Future Leaders in Advanced Materials Summer Symposium, UC Santa Barbara, Santa Barbara, CA, Aug. 2020.
5. **[Poster] J. V. Raj**, R. Valenzuela, and C. Marlow, "Electrical current characteristics of simulated carbon nanotube network field-effect transistors," APS Far West Section Meeting, Stanford University, Stanford, CA, Nov. 2019.

Teaching Assistant, UCLA

- Physics 1B: Oscillations, Waves, Electric and Magnetic Fields (Fall 2024, Spring 2025)
- Physics 1A: Mechanics (Winter 2025)
- Astronomy 115: Introduction to Galactic Astronomy (Winter 2023, Winter 2024)
- Physics 32: Mathematical Methods (Fall 2023)
- Physics 112: Thermal Physics (Spring 2023)

Mentoring and Outreach

- Research mentor to Pola V. Pietrzkowski (UCLA undergraduate) on how network topology affects physical learning (2026–present)
- Research co-mentor, with Mason A. Porter, to Emma Vega (UCLA undergraduate) on percolation in Erdős–Rényi, Barabási–Albert, and configuration-model networks (2025–present)
- Research co-mentor, with Mason A. Porter, to Xiaohan Sun (UCLA undergraduate) on curvature and effective resistance in networks generated from point patterns (2024–2025)
- Mentor to five undergraduate students, Critical MASS mentorship program, UCLA Marginalized Identities in Physics & Astronomy (MiPA) (2023–2026)
- Mentor to two incoming Ph.D. students, UCLA Physics Ph.D. Mentorship Program (2023–2026)

Honors and Awards

- Outstanding Teaching Award (twice), UCLA Physics and Astronomy (2022–2023 and 2023–2024)
- Future Leaders in Advanced Materials Internship Award (2020)
- William and Linda Frost Undergraduate Research Award, Cal Poly (2019)

Service

- Assisted with a referee report for *Interfaces and Free Boundaries* (2025)
- Administrator of the Porter research group's shared Linux compute server (2025–present)

Computational Skills

- **AI tools:** Claude Code and OpenAI Codex (custom skills, subagent pipelines); llama.cpp

Memberships

- American Physical Society (Member, 2018–present)
- Sigma Pi Sigma, physics honor society, Cal Poly (Vice President, 2020–2021)